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Machine agency attribution in human–AI interaction: developing and validating an analytical framework

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Abstract

As artificial intelligence (AI) increasingly reshapes human communication, the attribution of agency to machines has become a central scholarly concern. Yet there is little consensus on how users psychologically construe machine agency in human–AI interaction. To address this gap, we propose and validate an analytical framework that organizes prior conceptualizations into three phases: (a) perceptual phase (perception of agentic behavior), (b) inferential phase (inference of agentic minds), and (c) evaluative phase (judgment of agentic influence). Two survey studies develop measures for these phases, and an experiment tests their relationships. The results support a layered process: Perceived machine independence and goal-orientation (perceptual-phase variables) are directly associated with influential capacity judgment, and mental-state inference further strengthens this association. The framework clarifies how divergent conceptualizations of machine agency may reflect different phases of a broader attribution process and helps locate related claims at more precise levels of analysis.

Lay Summary

As artificial intelligence (AI) systems increasingly shape media experiences, a central question is what people mean when they describe a machine as "agentic." We propose a three-phase account of machine agency attribution: a perceptual phase, in which people perceive AI's behavior as independent and goal-directed; an inferential phase, in which they infer human-like mental states; and an evaluative phase, in which they judge that AI can make a meaningful difference. Across multiple studies, we find that these phases form a layered process: Perceiving AI as acting independently and toward specific goals can directly support judgments of its influence, whereas mental-state inference can further strengthen this link. These findings show how different meanings of machine agency can be understood as related phases of a broader attribution process in human–AI interaction.

When people describe a machine as "agentic," what are they actually referring to? As artificial intelligence (AI) systems increasingly assume active roles in shaping human communication, the attribution of agency to machines has become central to scholarship on human–AI interaction (HAII) and human–machine communication (HMC). However, the conceptual scope of machine agency attribution remains contested. Many scholars define or measure it through humanlike mental or cognitive capacities, such as intentionality (Sullins, 2006), consciousness or free will (Kang & Kim, 2020), or thought (Liao et al., 2023). Other work, by contrast, conceptualizes machine agency in terms of perceived system-afforded autonomous capabilities (Jia et al., 2012; Sundar, 2020) that are enacted or negotiated during HMC (Endacott, 2024; Gibbs et al., 2021), enabling machines to pursue goals and exert influence (Engen et al., 2016; Zafari & Koeszegi, 2018).

These differences suggest that prior research may not always be examining the same psychological aspect of machine agency attribution. For example, the mental-state account foregrounds higher-order mentalization, namely, the capacity to infer others' unobservable beliefs, desires, and intentions that guide their actions (Dennett, 1987; Rakoczy, 2022). By contrast, other accounts highlight perceptually grounded processing driven by specific technological affordances (Norman, 1999), which may not require mentalization. This conceptual variation raises two questions: What distinct psychological phases are implied by prior conceptualizations of machine agency? And how are these phases related within the broader process of machine agency attribution?

To address these questions, this study proposes a framework of machine agency attribution. The framework organizes prior accounts of machine agency into three analytically distinct phases: (a) perceptual phase (perception of agentic behavior, including the observation of agentic behavioral cues and the interpretation of agentic behavioral patterns); (b) inferential phase (inference of agentic minds); and (c) evaluative phase (judgment

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of agentic influence). Figure 1 presents the proposed framework. The following sections elaborate the variables for each phase and the relationships among them.

We examine this framework through two survey studies that develop corresponding measures and an experimental study that tests the proposed relationships among these phases. Rather than offering a definitive account of machine agency, the framework is intended to clarify how seemingly divergent conceptualizations of machine agency may reflect different phases of a broader attribution process in HAII. By empirically examining the relationships among the perceptual, inferential, and evaluative phases, the study clarifies how users construe agency in AI systems within a layered attribution process and provides a more precise basis for theorizing and operationalizing machine agency-related constructs across HAII and HMC research.

Attributing agency to machines: prior perspectives

The concept of agency has long been contested across intellectual traditions. Within Enlightenment thought, it was typically treated as a uniquely human attribute grounded in autonomous free will, and thus as a defining feature that distinguishes humans from other entities (Williams et al., 2021). Modern social theorists extended this view by defining agency more broadly as the human capacity to shape one's environment and life course (Bandura, 2008) and the power to act otherwise in any given sequence of conduct (Giddens, 1984). In media and communication research, these human-centered conceptions have been further elaborated in specific communicative contexts. Communicative agency, for instance, refers to audiences' capacity to exercise power through interpretive engagement in everyday communication (Ytre-Arne & Das, 2021), whereas in interactive media research, agency is often understood as users' capacity to act as communication sources by customizing media content for themselves and creating content for others (Sundar, 2007, 2020).

Influenced by these human-centered accounts, one prominent line of research on machine agency examines whether machines are perceived as possessing human-like mental or cognitive capacities. For instance, empirical studies have operationalized perceived machine agency in terms of users' beliefs about technological consciousness and free will (Kang & Kim,

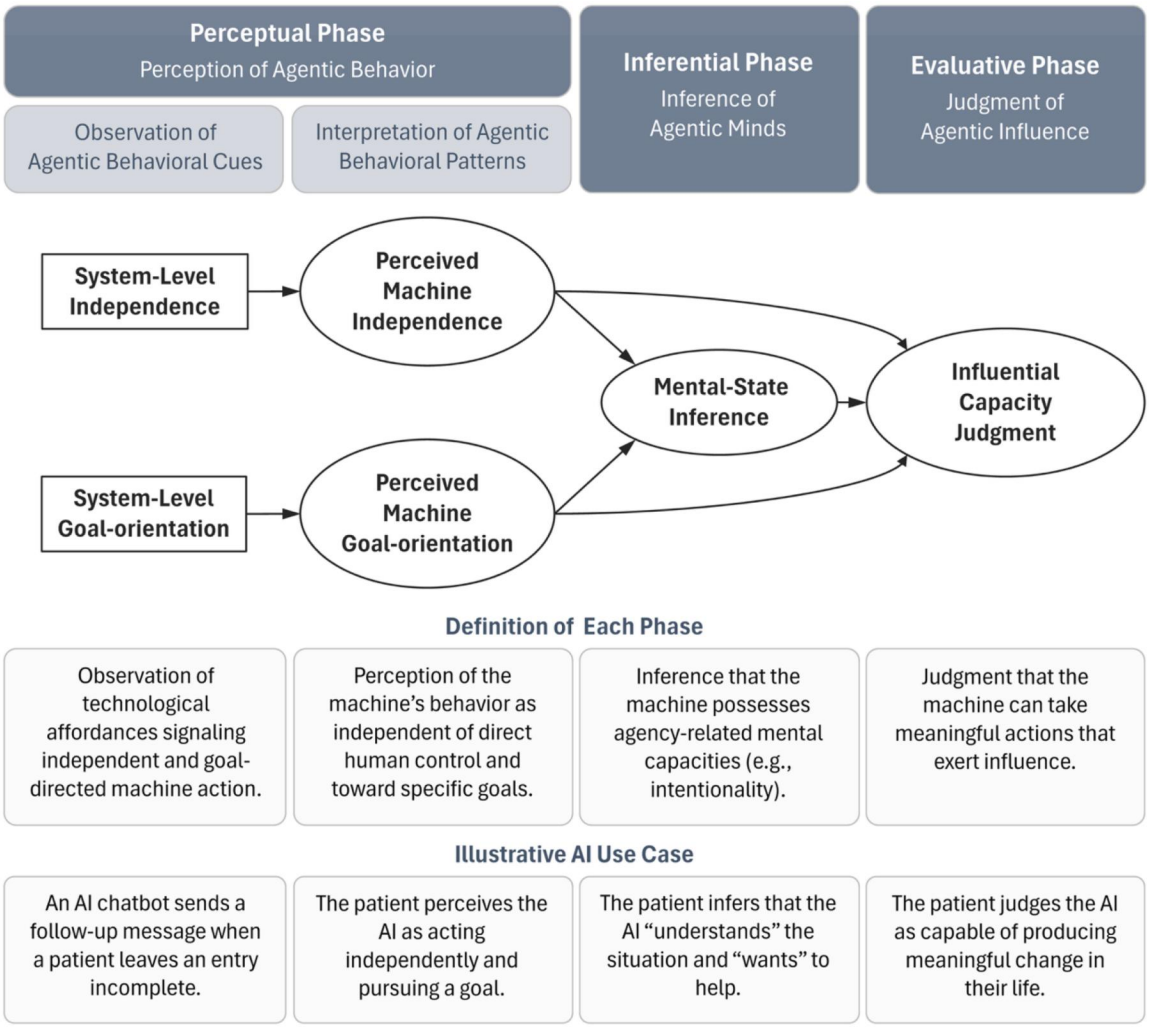


Figure 1. A layered framework of machine agency attribution in human–AI interaction.

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2020) or through dimensions of human agency that emphasize mental capacities such as thought and morality (Liao et al., 2023). Other work has manipulated machine agency using subjective language implying that a machine possesses its own feelings and beliefs (Pan et al., 2025).

Challenging these anthropocentric assumptions, Actor–Network Theory (ANT) reconceptualizes agency as the capacity to produce effects within sociotechnical networks, regardless of whether the acting entity is human, technological, or material (Latour, 1990). From this perspective, entities ranging from bacteria to water currents may be treated as agents insofar as they exert effects within relational systems (Anicker et al., 2024). Consistent with this view, machine agency has been conceptualized as the technological capacity to perform activities and exert influence within a given environment (Engen et al., 2016). In HMC, such agency is enacted by users through communication practice (Endacott, 2024), but it is also recursive in that human and machine agency become mutually constitutive over time (Gibbs et al., 2021). For example, users influence media by adjusting algorithmic parameters to suit their needs (Kang & Lou, 2022), while machines in turn shape human relationships and organizational power structures (Kirkwood et al., 2022).

Beyond mental and influence-based accounts, scholars have also explicated machine agency by focusing on how AI demonstrates task completion enabled by the system itself. In human–computer interaction, affordances generally refer to the actions a technology makes possible, whether in practice or in the user's perception (Norman, 1999). As intelligent technologies become increasingly capable of making autonomous selections and decisions, tasks that were once user-afforded are becoming system-afforded (Sundar, 2020). From this perspective, machine agency is defined as the user's perception that a technological system can automatically communicate and perform tasks on their behalf (Jia et al., 2012; Sundar et al., 2015).

Psychological phases of machine agency attribution

The inferential phase: inference of agentic minds

Although existing conceptualizations of machine agency vary, they can be understood as foregrounding different phases of a broader attribution process. Mental-capacity accounts, in particular, foreground an inferential phase centered on *mental-state inference*, whereby individuals interpret and predict an entity's behavior by attributing internal states such as beliefs, desires, and intentions (Dennett, 1987; Rakoczy, 2022). For example, when someone searches for their keys where they last saw them, unaware that another person has moved them, we interpret the behavior not as irrational but as guided by a false belief.

This human tendency to infer others' unobservable internal mental states, commonly referred to as the theory of mind (ToM; Premack & Woodruff, 1978), typically emerges between ages four and six (Wimmer & Perner, 1983) and extends beyond human–human interaction. Children, for instance, attribute mental states to social robots at developmental stages similar to those observed in human contexts (Zhang et al., 2019). Adults likewise often ascribe phenomenal consciousness to large language models such as ChatGPT (Colombatto & Fleming, 2024).

In the present framework, the content of this inferential phase is specified as the inference of agency-related mental capacities. Gray et al. (2007) found that people intuitively represent minds along two dimensions: agency, defined as the capacity for reasoned action, and experience, defined as the capacity for sensation and emotion. Because this study concerns machine agency attribution, the inferential phase is primarily concerned with mental capacities associated with the agency dimension, including intentionality, free will, consciousness, and self-awareness (Weisman et al., 2017). These inferred inner properties represent the kinds of mental capacities through which users may construe a machine as possessing an agentic mind.

The perceptual phase: perception of agentic behavior

Accounts centered on perceived technological affordances for autonomous task completion foreground a different phase of the attribution process, namely, a perceptual phase concerned with how users perceive what machines do rather than what machines think. Research within the Computers Are Social Actors (CASA) paradigm (Nass et al., 1994; Reeves & Nass, 1996) shows that even limited primitive cues related to humans, such as eye gaze (Admoni & Scassellati, 2017), can elicit social perceptions and responses, even when users recognize that machines do not possess human mental capacities. Building on this cue-based perspective, we conceptualize the perceptual phase as involving two sequential layers: the observation of agentic behavioral cues and the interpretation of agentic behavioral patterns.

Observation of agentic behavioral cues

In HAII, cues are perceivable interface indicators through which users detect and interpret underlying technological affordances (Sundar et al., 2015). Among the many cues that may appear in interaction, those signaling independent and goal-oriented machine action are especially central because they form part of the general understanding of what makes a system agentic (Wooldridge, 2009). On the one hand, *system-level independence* refers to a technological system's ability to initiate or execute operations on its own (Beer et al., 2014). Passive tools such as hammers exhibit minimal agency because they remain fully dependent on external force, whereas cooperative systems such as smart-home technologies exhibit a greater degree of agency because of self-coordinating functions (Rammert, 2008).

On the other hand, *system-level goal-orientation* refers to a technological system's designed capacity to pursue specific outcomes or complete tasks, making it another central yet distinct behavioral indicator of machine agency (Engen et al., 2016; Zafari & Koeszegi, 2018). Depending on the degree of human input, this capacity may be rule-based, as in spam filters or washing machines, or adaptive, as in machine-learning-based AI agents (Hayes-Roth, 1995; Hayes-Roth, 1985). In either case, system-level goal-orientation functions as a perceivable cue of machine agency because it makes the system's behavior appear organized around the pursuit of specific ends, thereby signaling a form of agentic action rather than undirected or purely mechanical response.

Although some prior work treats independence and goal-orientation as a single definitional bundle of machine agency (e.g., Zafari & Koeszegi, 2018), we conceptualize them as distinct but

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related dimensions because they signal different aspects of machine affordances. A system may appear goal-oriented while remaining weakly independent, as in rigid rule execution under strong external constraints, whereas more adaptive systems may exhibit both. Distinguishing the two therefore allows for a more precise account of how agentic system cues are represented in HAII.

Interpretation of agentic behavioral patterns

At the interpretive level of the perceptual phase, the key constructs are perceived machine independence and goal-orientation. *Perceived machine independence* refers to the extent to which users perceive a machine's actions as occurring without direct human control or external intervention, whereas *perceived machine goal-orientation* refers to the extent to which users perceive a machine's actions as purposefully directed toward specific objectives or plans rather than as vague, random, or arbitrary. Both perceptions are elicited by the corresponding system-level cues discussed earlier. Unlike system-level affordances themselves, these constructs capture users' interpretations of how a machine behaves in interaction rather than the affordances objectively built into the system, and such interpretations are often more consequential in human–computer interaction (Norman, 1999).

At the same time, because mentalization is an ingrained human tendency, users may move from interpreting behavioral cues to inferring mental states. This overlap creates what Rammert (2014, p. 5) calls "the practical fiction of technology in action," in which terms such as autonomy, volitional control, or intentionality are used to describe machine functioning without clearly distinguishing observable behavior from inferred mentality. To preserve the conceptual boundary between the CASA-based perceptual phase and the ToM-based inferential phase, we define perceived machine independence and perceived machine goal-orientation in explicitly behavioral rather than mentalistic terms.

Unlike human autonomy or free will, which ToM-based accounts define in terms of internal volition (Ryan & Deci, 2000), perceived machine independence concerns the extent to which a system appears to act without direct human intervention during interaction (Beer et al., 2014). For example, manual-transmission cars require continuous human operation and are therefore perceived as less independent, whereas self-driving vehicles that initiate actions without continuous driver input are perceived as more independent. By contrast, human intentionality, from a ToM perspective, presupposes beliefs and desires that motivate action (Baldwin & Baird, 2001; Malle & Knobe, 1997), whereas perceived machine goal-orientation can be formed without any commitment to inner mental states, based instead on observable input–output regularities (cf. Rosenblueth et al., 1943). For example, when a vehicle deviates from its planned route, passengers typically question whether the system is successfully pursuing its intended destination rather than attributing malicious intent to it. Together, these examples show that perceiving machine behavior as independent and goal-directed does not by itself imply the attribution of humanlike autonomy or intentionality unless users move beyond behavioral interpretation to explicit mental-state inference.

The evaluative phase: judgment of agentic influence

Influence-oriented accounts foreground a third phase of the attribution process, namely, an evaluative phase. In this phase, users make an *influential capacity judgment* about whether a machine possesses the power to take meaningful actions that produce effects, exert influence, or shape interaction outcomes. Rather than asking whether a machine possesses humanlike mental capacities or whether its behavior appears independent and goal-directed, this phase asks whether the attributed agency is consequential: Can the system actually affect users, decisions, or the broader trajectory of interaction?

Across diverse traditions of agency theorization, whether human or nonhuman, biological or material, the capacity to produce meaningful effects consistently appears as a core criterion of agency. Here, "meaningful" refers to discernible consequence rather than mere activity (Murray, 1998). An AI system may generate multiple suggestions, for example, but if none of them shape the user's actions or the course of interaction, its influence on outcomes remains difficult to discern. This logic appears in prior conceptualizations of agency as the power to shape one's immediate environment (Bandura, 2006), the nature of media content (Sundar et al., 2015), and, more broadly, the reconstruction of society (Giddens, 1984).

This evaluative phase is critical because it provides the consequential criterion of machine agency attribution. Whereas the perceptual and inferential phases specify how agency is recognized and interpreted, the evaluative phase determines whether that attributed agency is judged to matter in practice. In this framework, the evaluative phase serves as the culminating endpoint of machine agency attribution.

Process of machine agency attribution: a layered framework

If these phases locate machine agency attribution at different psychological levels, a further question concerns how they relate to one another, as no single phase fully captures how machine agency attribution unfolds in HAII. The inferential phase alone cannot account for agency attribution when users regard core mental attributes, such as consciousness or free will, as ontologically absent, especially given that mentalization in HMC can be fragile and easily disrupted (Banks, 2020) and that people rarely ascribe to machines mental capacities fully comparable to those of humans (Gray et al., 2007; Weisman et al., 2017). At the same time, the perceptual phase, while offering a CASA-based account grounded in cue-triggered behavioral perceptions, cannot fully substitute for the inferential phase, because ToM reflects an ingrained tendency to make sense of complex machine behavior (Rakoczy, 2022) and to sustain socially meaningful relations with nonhuman agents (Lee & Hahn, 2024). The evaluative phase, in turn, is needed because agency attribution ultimately concerns not only whether a machine appears agentic, but also whether that attributed agency is judged to matter in practice. These phases are therefore theorized here as parts of a layered process rather than as separate accounts.

Prior literature further suggests that the perceptual, inferential, and evaluative phases may be hierarchically related. In social cognitive theory, mental capacities associated with agency,

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such as intentionality, have long been regarded as central to understanding an entity as capable of purposive action and meaningful influence (Bandura, 2001). Accordingly, the more users infer that a machine possesses such mental states, the more likely they may be to judge it as capable of exerting meaningful influence. We therefore propose:

H1: *Mental-state inference will be positively associated with influential capacity judgment.*

The perceptual phase may also provide an important basis for the inferential phase. Leslie's (1994) tripartite theory of agency, for example, proposes a three-level hierarchy involving mechanical properties, actional understanding, and mental-state reasoning. These levels parallel users' observation of agentic behavioral cues, their interpretation of agentic behavioral patterns, and their inference of agentic minds, respectively. The anthropomorphism framework proposed by Kühne and Peter (2023) points in a similar direction by suggesting that observable technological features facilitate mind perception and humanlike attributions. Together, these accounts imply that mental-state inference is often grounded in behavioral interpretation rather than emerging spontaneously.

Empirical evidence supports this perceptual–inferential linkage. Mental-state attribution to robots is primarily driven by observed behavioral patterns (Thellman et al., 2022). When a robot behaves independently of external regulation, such as by acting unpredictably (Eyssel et al., 2011) or violating norms (Korman et al., 2019), or in overtly goal-directed ways, such as moving toward a displaced object (Yamaji et al., 2010), people become more likely to attribute mental states to it. We therefore propose:

H2: *Perceived machine independence, triggered by system-level independence cues, will be positively associated with mental-state inference.*
H3: *Perceived machine goal-orientation, triggered by system-level goal-orientation cues, will be positively associated with mental-state inference.*

At the same time, prior literature suggests a theoretical tension regarding the role of the inferential phase in linking the perceptual and evaluative phases. From the CASA perspective, perceiving computers as social entities is "not the result of conscious beliefs that computers are human or humanlike" (Nass et al., 1994, p. 72). Instead, social responses and perceptions are automatically and mindlessly triggered by minimal cues (Nass & Moon, 2000; Xu et al., 2022), suggesting that the cue-triggered behavioral perceptions emphasized in the present framework may already be sufficient to support judgments of agentic influence.

By contrast, mind-perception accounts assign a more central role to inferred mentality in evaluating agentic capability. In both Gray et al.'s (2007) agency-experience framework and Weisman et al.'s (2017) body–heart–mind account, capacities such as free will, consciousness, and intentionality are treated as important bases for interpreting and predicting an entity's behavior as the product of an agentic mind. These competing views raise a key question: Does the inference of agentic minds fully mediate the relationship between cue-triggered behavioral perceptions and judgments of agentic influence, or do these perceptual judgments also exert a direct effect? We therefore ask:

RQ: *Will mental-state inference fully mediate the relationship between perceived machine independence and goal-orientation and influential capacity judgment, or will these relationships be only partially mediated?*

Study 1: development of the perceptual-phase measures

Although perceived machine independence and goal-orientation have been discussed conceptually in prior work, neither has been developed into a psychometrically validated measure. To address this gap, Study 1 developed preliminary scales for these two perceptual-phase constructs. To avoid contamination across attribution phases, the evaluative-phase measure, influential capacity judgment, was developed separately in Study 2. All resulting measures were subsequently refined and validated within the framework tested in Study 3.

Preliminary item generation

Scale development began with the generation of a preliminary item pool. To maximize content validity, items should be developed deductively from a construct's theoretical definition rather than generated inductively without such grounding (Cronbach & Meehl, 1955; Hinkin, 1998). Because this study offers explicit definitions of perceived machine independence and perceived machine goal-orientation, and distinguishes them from adjacent mental-state constructs, we generated items directly from these definitions, consistent with established scale-development guidance (Carpenter, 2018; Hinkin, 1998; MacKenzie et al., 2011).

As defined earlier, perceived machine independence refers to perceiving a machine's actions as occurring without direct human control or intervention, whereas perceived machine goal-orientation refers to perceiving a machine's actions as purposefully directed toward specific objectives rather than as vague or arbitrary. Based on these definitions, we created eight items, four for independence (IN1a–IN4a) and four for goal-orientation (GO1a–GO4a).

Because conceptually derived items may contain terminology unfamiliar to lay respondents, we also generated a parallel set of eight plain-language items (IN1b–IN4b; GO1b–GO4b) designed to preserve the intended meaning while improving accessibility. All items were refined through feedback from a focus group of general AI users and scholars ($N = 9$). This two-tiered item-generation approach ensured that the preliminary items were both conceptually grounded and readily comprehensible. Details of items are provided in Supplementary Table 1, see online supplementary material.

Participants, stimulus, and procedure

A quota sample of U.S. adults ($N = 355$) was recruited via Prolific to approximate the population distribution of age and sex (see Supplementary Table 2, see online supplementary material for

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demographics). Participants viewed a JavaScript-powered interactive stimulus built on Wix that simulated an HAII scenario (see Supplementary Figure 1, see online supplementary material). The stimulus provided a concrete, context-rich display of agentic machine behavior, allowing participants to evaluate the items based on observed interaction cues rather than imagination alone.

The scenario depicted a user asking an AI system to generate an email explaining a class absence to a course instructor. This task was selected because email generation is a common AI use case and the classroom setting is likely familiar to most participants, thereby enhancing mundane realism by resembling everyday experience (Aronson et al., 1998). The interaction followed a pre-scripted dialog sequence (see Supplementary Table 3, see online supplementary material).

After providing informed consent and reading the cover story, participants viewed the stimulus and then completed a questionnaire assessing perceptions of the AI's independence and goal-orientation, followed by the additional measures described below.

Measures

Participants rated the initial perceived machine independence and goal-orientation items on a 7-point Likert scale ranging from 1 (*not at all*) to 7 (*very much so*). Item order was randomized to reduce response bias.

To ensure data quality, the questionnaire included attention checks (e.g., "For this question, please select 'neutral' to demonstrate your attention") and stimulus-recall questions (e.g., "Please recall what is the purpose of the email"), following established methodology and practice (Kung et al., 2018; Wu et al., 2024). Participants who failed both checks were excluded from analysis. Of the 372 respondents, 17 were removed, yielding a final sample of 355, which exceeded recommended minimums for factor analysis (Carpenter, 2018).

An open-ended question at the end of the questionnaire asked whether respondents had encountered any difficulty interpreting the scale items. No major interpretation difficulties were reported.

Exploratory factor analysis and items purification

Exploratory factor analysis (EFA) was conducted to examine the hypothesized factor structure and refine the preliminary items. As preliminary diagnostics, Bartlett's test of sphericity was significant for both the independence items, $\chi^2(28) = 1{,}342.81$, $p < .001$, and the goal-orientation items, $\chi^2(28) = 1{,}231.20$, $p < .001$. Kaiser–Meyer–Olkin (KMO) values also indicated adequate sampling for factor analysis ($KMO_{Independence} = 0.904$ and $KMO_{Goal\text{-}orientation} = 0.903$; Shrestha, 2021).

We then conducted a joint EFA on the full initial pool of items measuring perceived machine independence and perceived machine goal-orientation (see Supplementary Table 4, see online supplementary material). Although parallel analysis (see Supplementary Figure 2A, see online supplementary material) suggested a three-factor solution, the third factor was weak, lacked clear theoretical interpretability, and showed substantial cross-loadings (see the "3-Factor" columns in Supplementary Table 4, see online supplementary material). We therefore retained the two-factor solution, which yielded a cleaner, more interpretable structure (Costello & Osborne, 2005).

Item retention followed established criteria (Carpenter, 2018; Worthington & Whittaker, 2006). Items were retained if they (a) did not show substantial cross-loadings on multiple factors, defined here as loadings greater than 0.32 given the sample size (Hair et al., 2010); (b) loaded more strongly than the matched item in the paired wording alternative; and (c) preserved at least four items per factor to maintain both conceptual coverage and parsimony. Based on these criteria, we retained IN1a, IN2a, IN3b, and IN4a for perceived machine independence, and GO1a, GO2a, GO3b, and GO4a for perceived machine goal-orientation. All retained items loaded above 0.562, exceeding the 0.32 threshold for this sample size (Carpenter, 2018; Hair et al., 2010).

Study 2: development of the evaluative-phase measure

Study 2 aimed to develop and refine a scale for the influential capacity judgment.

Preliminary item generation

Following the deductive item-generation strategy used in Study 1, we developed eight items (IC1–IC8) directly from our conceptualization of influential capacity judgment as the evaluation that machines possess the power to take meaningful actions that produce effects, exert influence, or bring about change. Details of items are provided in Supplementary Table 5, see online supplementary material.

Participants, stimulus, and procedure

Another quota sample of U.S. adults ($N = 245$) was recruited via Prolific (see Supplementary Table 6, see online supplementary material for demographics). The stimulus design mirrored Study 1 but replaced the class-absence scenario with a meeting-rescheduling request to introduce variation and reduce scenario-specific bias (see Supplementary Table 7, see online supplementary material for pre-scripted dialog). The procedure otherwise followed Study 1.

Measures

Items were rated on a 7-point Likert scale (1 = *not at all*, 7 = *very much so*), with item order randomized. As in Study 1, the questionnaire included attention checks, stimulus-recall questions, and an open-ended item-comprehension question. Of the 251 initial respondents, six were excluded for failing both attention and recall checks, yielding a final sample of 245, which exceeded the recommended minimum for EFA (Carpenter, 2018). No major item-interpretation difficulties were reported.

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Exploratory factor analysis and items purification

Bartlett's test of sphericity was significant, $\chi^2(28) = 1{,}556.67$, $p < .001$, and the KMO value (.866) exceeded the recommended 0.60 threshold (Shrestha, 2021), indicating that the data were suitable for factor analysis. Although parallel analysis suggested two factors (Supplementary Figure 2B, see online supplementary material), as shown in Supplementary Table 8, see online supplementary material, the second factor was defined only by IC5 and IC6, falling below the recommended minimum of three indicators for retaining a factor as a separate construct (Carpenter, 2018). Moreover, when these items were estimated in a one-factor solution, their loadings dropped substantially (IC5 = 0.642; IC6 = 0.630), suggesting weaker integration with the primary factor defined by the remaining items.

Inspection of the item content clarified this divergence. Unlike the other items, which assessed influential capacity judgment as a micro-level capacity relevant to the specific AI-use scenario, IC5 and IC6 referred to broader, world-transforming effects (e.g., "transcend its environment," "reconstruct the world"). Taken together, these patterns suggested that IC5 and IC6 captured a different dimension from the intended construct, and both items were therefore removed.

IC2 was also excluded. Although its loading in the one-factor solution was acceptable (0.741), its substantially weaker loading in the two-factor solution (0.527), relative to the retained items, suggested lower stability and weaker alignment with the core factor. The refined influential capacity judgment scale therefore consisted of five items: IC1, IC3, IC4, IC7, and IC8. All retained items showed strong loadings in the one-factor solution (≥0.770), exceeding the recommended 0.40 cutoff for a sample size of 245 (Hair et al., 2010).

Study 3: measurement validation and framework verification

Study 3 had two objectives: (a) to confirm the measurement model and assess the reliability and validity of the refined scales for the perceptual-phase constructs, perceived machine independence and goal-orientation, as well as the evaluative-phase construct, influential capacity judgment; and (b) to test the structural relationships among the variables in the proposed framework.

Experimental design, stimuli, participants, and procedure

According to the proposed framework (Figure 1), system-level independence and goal-orientation are expected to cue perceived machine independence and goal-orientation, which, in turn, are expected to relate to mental-state inference and influential capacity judgment. To activate this psychological process, we implemented a 2 (system-level independence cues: absence vs. presence) × 2 (system-level goal-orientation cues: absence vs. presence) between-subjects experiment.

This design offered two advantages. First, experimentally manipulating the presence or absence of system-level agentic behavioral cues allowed framework validation to be based on varied rather than static stimuli. Second, the factorial design enabled us to test whether the perceived machine independence and goal-orientation scales corresponded to their dummy-coded experimental manipulations, thereby establishing veridical validity (or veridicality; MacKenzie et al., 2011), that is, whether the scales accurately captured the phenomena they were designed to measure.

Four webpages were created using the Study 1 approach. The scenario, adapted from AI self-disclosure research (Meng & Dai, 2021; Mou et al., 2023), depicted an emotionally supportive exchange in which a user asked an AI how to comfort a third party after unemployment (see Supplementary Table 9, see online supplementary material for pre-scripted dialogue). This scenario was selected to provide a socially meaningful, emotionally rich context for evaluating the measured constructs.

Using the same sampling source as in Studies 1 and 2, participants were randomly assigned to one of the four experimental conditions. Of the 244 respondents, 16 failed both attention and recall checks and were excluded, yielding a final sample of 228 (see Supplementary Table 10, see online supplementary material for demographics). Participants viewed the interactive animation corresponding to their assigned condition and then completed the questionnaire.

Experimental manipulations

In the absence of system-level independence cues, the AI's responses were passive and user-controlled: It generated output only after receiving explicit prompts (e.g., "I instruct you to generate the message now"). In the presence of system-level independence cues, the AI initiated and executed a sequence of steps immediately after receiving the initial request, without requiring further user input.

In the presence of system-level goal-orientation cues, the AI's responses included specific goals (e.g., "I will write a message to acknowledge his feelings") and concrete action plans (e.g., "demonstrating (a) the difficulty of the situation, (b) understanding the pain of unemployment, and (c) validating his emotions"). In the absence of these cues, the AI offered only general plans (e.g., "I will write something for him") without elaborating specific steps.

To ensure that the manipulations reflected only AI-driven variation, users did not provide task- or goal-specific instructions in any condition. Example dialog scripts by manipulation condition are provided in Supplementary Table 9, see online supplementary material.

Measures and data analysis plan

Perceived machine independence, *perceived machine goal-orientation*, and *influential capacity judgment* (independence: $\alpha = 0.93$, $M = 3.47$, $SD = 1.72$, Skewness = 0.16; goal-orientation: $\alpha = 0.76$, $M = 4.75$, $SD = 1.30$, Skewness = −0.64; influence: $\alpha = 0.95$, $M = 4.37$, $SD = 1.65$, Skewness = −0.68) were assessed using the retained items from Studies 1 and 2, rated on the 7-point Likert scale.

Mental-state inference ($\alpha = 0.89$, $M = 2.33$, $SD = 1.04$, Skewness = 0.56) was measured with six items adapted from

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established agency-related mind perception scales (Gray et al., 2007; Weisman et al., 2017). Example items included "The machine is unconscious———conscious," "cannot think———does think," and "has no free will———does have free will," rated on a 5-point semantic-differential scale.

In addition, apart from *socio-demographic variables* (sex, age, education level, annual household income), we included participants' pre-existing *algorithm knowledge level* ($M = 8.15$, $SD = 1.50$, Skewness = −0.93) and AI-related *power usage level* (α = 0.83, $M = 4.45$, $SD = 1.19$, Skewness = −0.22) as control variables, given that users' knowledge and competence of AI usage can affect their overall perceptions toward AI. Algorithm knowledge was measured using 10 true-or-false items assessing objective knowledge of algorithms (e.g., "I can influence algorithms with my Internet usage behavior" [true]) adopted from Dogruel et al. (2022). AI-related power usage was measured with five items (e.g., "I make good use of various features of AI technology") on 7-point Likert-type scales adapted from Kang and Shin (2016).

The complete questionnaire is provided in Supplementary Table 11, see online supplementary material, descriptive statistics by experimental condition are reported in Supplementary Table 12, see online supplementary material, and intercorrelations are reported in Supplementary Table 13, see online supplementary material. Shapiro–Wilk tests indicated significant deviations from normality for all variables ($p < .05$), suggesting that the assumption of multivariate normality required for conventional covariance-based SEM may not hold. Given these distributional characteristics, together with the study's emphasis on theoretical model development, we employed variance-based partial least squares SEM (PLS-SEM) in SmartPLS 4.1 (Ringle et al., 2024). PLS-SEM is less sensitive to violations of normality and is particularly well-suited to theory extension and development (Hair et al., 2021). We used bootstrapping with 10,000 resamples to assess the measurement model and test the structural model.

Results

Measurement model confirmation

The measurement model included perceived machine independence and perceived machine goal-orientation as independent variables, mental-state inference as the mediator, and influential capacity judgment as the dependent variable. Control variables and experimental-manipulation dummies were excluded at this stage so that the construct structure could be evaluated in a pure measurement model (Brown, 2015; Kline, 2011).

The PLS-SEM model showed good fit (SRMR = 0.064), below the recommended 0.10 threshold (Hu & Bentler, 1999; Ringle et al., 2024). A supplementary covariance-based confirmatory factor analysis using maximum likelihood estimation yielded

Table 1. Measurement model confirmation for the key constructs (Study 3).

Perceived machine independence	**SL**	**α**	**CR**	**AVE**
[IN1a] The machine acted independently without relying on others.	0.934	0.934	0.953	0.834
[IN2a] The machine made its own decisions without external control.	0.917			
[IN3b] The machine's actions resulted from itself rather than other entities.	0.887			
[IN4a] The machine organized its actions by itself without other influence.	0.915			
Perceived machine goal-orientation	**SL**	**α**	**CR**	**AVE**
[GO1a] The machine acted purposefully rather than merely behaving.	0.829	0.755	0.844	0.582
[GO2a] The machine's actions were task-oriented.	0.542			
[GO3b] The machine had goals behind its actions.	0.818			
[GO4a] The machine's actions were part of a plan.	0.823			
Mental-state inference (adapted from established scales)	**SL**	**α**	**CR**	**AVE**
[MS1] "The machine is unconscious"———"The machine is conscious"	0.850	0.893	0.918	0.653
[MS2] "The machine cannot think"———"The machine does think"	0.818			
[MS3] "The machine has no free will"———"The machine does have free will"	0.795			
[MS4] "The machine has no desires"———_"The machine does have desires"	0.760			
[MS5] "The machine is not self-aware"———_"The machine is self-aware"	0.861			
[MS6] "The machine has no intentionality"———_"The machine does have intentionality"	0.757			
Influential capacity judgment	**SL**	**α**	**CR**	**AVE**
[IC1] The machine is an agentic entity with the power to produce effects.	0.930	0.953	0.964	0.844
[IC3] The machine is an agentic entity with the power to achieve outcomes.	0.910			
[IC4] The machine is an agentic entity with the power to bring about change.	0.870			
[IC7] The machine is an agentic entity with the power to take meaningful actions.	0.935			
[IC8] The machine is an agentic entity with the power to make a difference.	0.947			

Note. Results are from the pure measurement model and therefore exclude experimental manipulation dummies and control variables. α = Cronbach's alpha; AVE = average variance extracted; CR = composite reliability; SL = standardized (outer) loading.

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comparable fit, $\chi^2(147) = 303.99$, $p < .001$, $\chi^2/df = 2.07$, CFI = 0.95, TLI = 0.95, and RMSEA = 0.068, further supporting the adequacy of the measurement model.

As shown in Table 1, all items loaded significantly ($p < .001$), with standardized loadings ranging from 0.542 to 0.947 and exceeding the 0.40 benchmark for a sample size of 228 (Hair et al., 2010). Reliability was high, with Cronbach's alpha values of 0.934 for independence, 0.755 for goal-orientation, 0.893 for mental-state inference, and 0.953 for influential capacity judgment, and composite reliability values of 0.953, 0.844, 0.918, and 0.964, respectively, all above the 0.70 criterion (Shrestha, 2021). Convergent validity was supported, with AVE values of 0.834, 0.582, 0.653, and 0.844, all exceeding the 0.50 benchmark. Discriminant validity (Table 2) was also established, as the square root of each construct's AVE exceeded its correlations with the other constructs (Fornell & Larcker, 1981).

Table 2. Discriminant validity of the key constructs (Study 3).

	1	2	3	4
1. Perceived machine independence	**0.913**			
2. Perceived machine goal-orientation	.398***	**0.763**		
3. Mental-state inference	.519***	.491***	**0.808**	
4. Influential capacity judgment	.468***	.531***	.550***	**0.919**

Note. Bold diagonal values indicate the square root of AVE; off-diagonal values are inter-construct correlations.
*** $p < .001$.

Structural model evaluation

To test the hypothesized structural relationships in the proposed framework, we estimated a consistent PLS-SEM including the dummy-coded experimental manipulations of system-level independence cues (0 = absence, 1 = presence) and system-level goal-orientation cues (0 = absence, 1 = presence) as independent variables. Sociodemographic variables (sex: 0 = male, 1 = female; age; education; income), algorithm knowledge level, and AI-related power usage level were included as control variables.

The structural model demonstrated acceptable fit (SRMR = 0.086), below the 0.10 threshold (Hu & Bentler, 1999; Ringle et al., 2024). As shown in Figure 2 and Supplementary Table 14, see online supplementary material, the experimental manipulations significantly increased the corresponding perceptual-phase variables: System-level independence cues increased perceived machine independence ($\beta = 0.622$, $t = 5.239$, $p < .001$), and system-level goal-orientation cues increased perceived machine goal-orientation ($\beta = 0.270$, $t = 2.271$, $p = .023$). These results support the veridical validity of the two perceptual-phase measures.

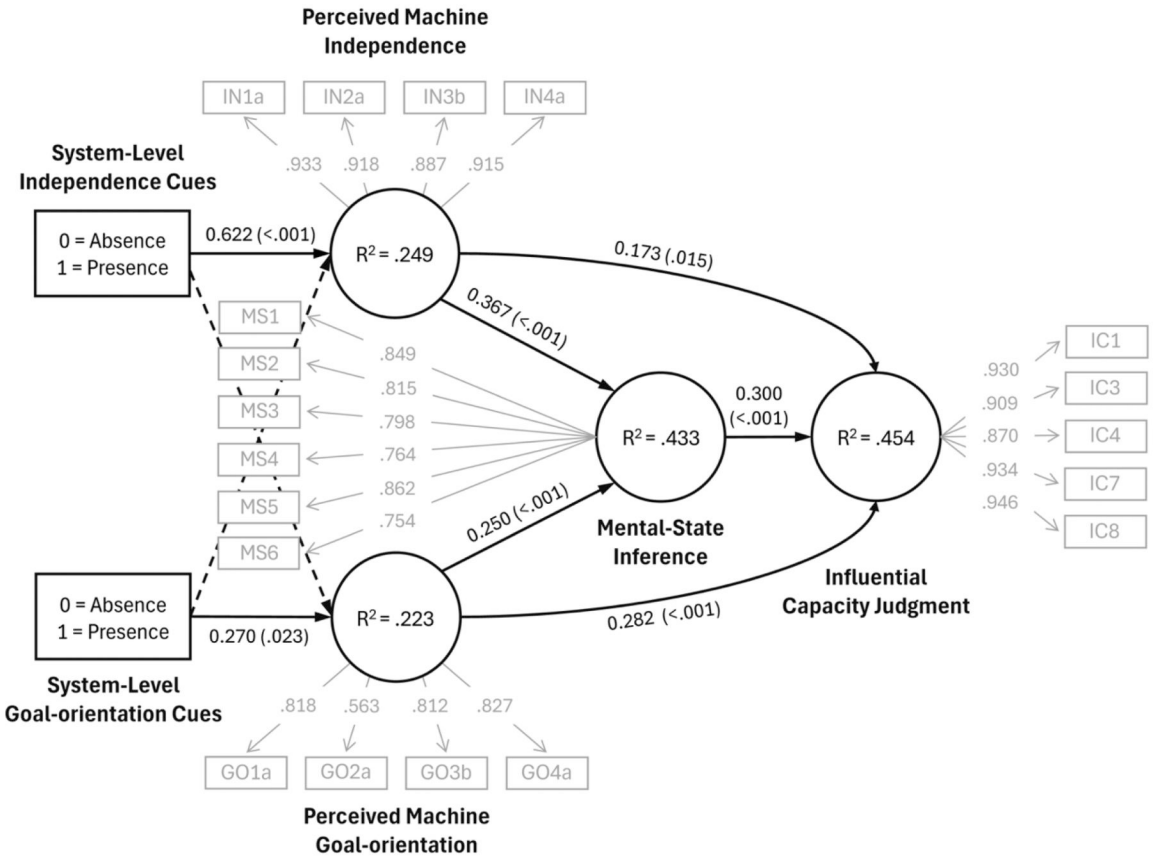


Figure 2. Structural model results for Study 3. *Note*. Solid lines indicate significant effects; dashed lines indicate nonsignificant effects. Control variables are omitted for visual clarity and reported in Table S14.

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Turning to the hypothesized framework relationships, perceived machine independence was positively associated with mental-state inference (β = 0.367, t = 5.745, $p < .001$), and perceived machine goal-orientation was also positively associated with mental-state inference (β = 0.250, t = 4.447, $p < .001$), supporting H2 and H3, respectively. Mental-state inference was positively associated with influential capacity judgment (β = 0.300, t = 4.201, $p < .001$), supporting H1.

Regarding the research question, both perceived machine independence (β = 0.173, t = 2.440, p = .015) and perceived machine goal-orientation (β = 0.282, t = 3.745, $p < .001$) were positively associated with influential capacity judgment. In addition, mental-state inference partially mediated the relationship between perceived machine independence and influential capacity judgment (β = 0.110, t = 3.225, p = .001), as well as the relationship between perceived machine goal-orientation and influential capacity judgment (β = 0.075, t = 3.197, p = .001). These findings indicate that perceiving a machine as independent and goal-oriented can directly support influential capacity judgment, without requiring mental-state inference as the sole pathway.

Discussion

This study develops a process-oriented account of machine agency attribution in HAII and HMC. Rather than treating machine agency as a single, self-evident construct, the proposed framework distinguishes among users' perceptions of agentic machine behavior, inferences of agentic minds, and judgments of agentic influence. Across three studies, the findings suggest that these phases are conceptually distinct yet psychologically related. Below, we discuss the conceptual, theoretical, and methodological implications of this framework, followed by its limitations and directions for future research.

Conceptual implications

A central conceptual contribution of this study is to clarify what different claims about "machine agency" psychologically refer to in users' experience of HAII. Prior research has used the term to describe related but non-equivalent phenomena, including perceived autonomous technological capabilities (Jia et al., 2012; Sundar et al., 2015), inferred human-like mental capacities (Kang & Kim, 2020; Sullins, 2006), and the consequential power to accomplish goals (Engen et al., 2016; Zafari & Koeszegi, 2018). The present framework organizes these accounts by locating them within three psychological phases of machine agency attribution: perceptual, inferential, and evaluative.

This distinction matters because some apparent disagreements in the literature may reflect differences in the psychological level at which machine agency is being theorized, rather than direct conceptual contradiction (cf. Comrey, 1988). From this perspective, the value of the framework lies not in imposing a single definition of machine agency, but in situating different conceptualizations within a shared attribution framework. This, in turn, may help future research specify more clearly which phase of machine agency attribution is under examination, thereby improving conceptual precision.

The findings further suggest that these phases are not merely taxonomic categories, but parts of a layered attribution process in which cue-triggered behavioral perceptions provide a lower-order basis for higher-order mental-state inference and subsequent judgments of agentic influence. This process-oriented view extends prior work by encouraging researchers to consider not only which aspect of machine agency attribution they are studying, but also how that aspect relates to other phases in the broader process. More broadly, this account suggests that what users mean by "machine agency" is not a single psychological referent, but a layered attribution that takes shape across perceptual, inferential, and evaluative phases.

Theoretical implications

The study also contributes to theorizing how CASA- and ToM-based processes may jointly shape judgments of agentic influence in HAII. A longstanding debate concerns whether anthropomorphism, often understood as the attribution of ToM-based mental states to machines (Kühne & Peter, 2023), is necessary for users to regard machines as agentic social entities capable of exerting meaningful influence. One line of work suggests that such judgments can emerge from minimal cues without genuine anthropomorphic inference (Nass & Moon, 2000), whereas other perspectives argue that anthropomorphism remains central, with primary cues eliciting "mindless" anthropomorphism and secondary cues eliciting more "mindful" forms (Lombard & Xu, 2021).

Our findings suggest that users need not infer a machine mind to judge a machine to be capable of meaningful agentic influence. Cue-triggered perceptions of machine independence and goal-orientation were directly associated with stronger judgments of agentic influence, but were also associated with heightened mental-state inference, which further strengthened those judgments. This pattern suggests that mental-state inference is better understood not as a prerequisite for influential capacity judgment, but as a higher-order process that can amplify judgments already grounded in perceived agentic behavior.

More broadly, the direct association between behavioral perceptions and influential capacity judgments highlights a limitation of purely ToM-based accounts, especially in contexts where anthropomorphic mentalization is weak, unstable, or absent. In HAII, mental-state inference may be unstable: It may be weakened by reflection on whether machines can genuinely possess minds (Banks, 2020) or triggered by superficial cues only loosely related to actual system capacity, such as humanlike appearance (Broadbent et al., 2013). In this sense, users' judgments that a machine can make a meaningful difference in HAII do not always require anthropomorphic inference, but may instead rest simply on whether its behavior appears capable of accomplishing specific goals without direct human control.

Methodological implications

Methodologically, the present study offers an initial step toward specifying a nomological structure for machine agency attribution. By outlining how perceptual-, inferential-, and evaluative-phase variables relate to one another, the framework provides a

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reference for the validation of machine agency-related constructs and the specification of models in HAII and HMC.

Crucially, the findings suggest that influential capacity judgment may function as a theoretically relevant downstream criterion for constructs in the perceptual and inferential phases. As an evaluative judgment concerning whether a machine can exert discernible influence on the interaction, it is positioned downstream from perceived machine independence, perceived machine goal-orientation, and mental-state inference, while remaining conceptually separable from them. Accordingly, the extent to which these upstream constructs relate to influential capacity judgment in theoretically predicted ways provides evidence for their construct validity within the proposed nomological structure (Cronbach & Meehl, 1955).

In addition, the study introduces preliminary scales for perceived machine independence, perceived machine goal-orientation, and influential capacity judgment. These measures move research beyond binary claims about whether machines are seen as agentic and enable more precise analysis of how specific machine features shape users' perceptions, inferences, and evaluations.

Limitations and future directions

Several limitations qualify these conclusions. First, empirical validation of the proposed framework remains preliminary, and the framework itself should be regarded as provisional rather than definitive. Although the experiment provided causal evidence that system-level independence and goal-orientation cues shape corresponding psychological phases, the downstream causal chain linking perceived machine independence, perceived machine goal-orientation, mental-state inference, and influential capacity judgment remains to be established. Moreover, because the framework and variables examined here are grounded in the machine-agency literature available at the time of writing, they may require refinement, expansion, or reorganization as that literature evolves. The framework should therefore be viewed as an empirically grounded intervention in an ongoing theoretical conversation rather than a final settlement.

Second, although multifactor solutions occasionally emerged during scale development, they either lacked clear interpretability or yielded factors with too few indicators to satisfy recommended psychometric standards (Carpenter, 2018; Costello & Osborne, 2005). Future research should therefore examine whether perceived machine independence, perceived machine goal-orientation, and influential capacity judgment contain meaningful subdimensions not captured by the present measures.

Third, the Study 3 manipulations focused on specific forms of independence and goal-orientation cues, and the scenarios examined here, email generation and emotional support, represent only a limited subset of AI use cases. Future research should therefore test the framework across a broader range of agentic machine behaviors, tasks, stakes, and interaction modalities in order to clarify its boundary conditions and broader applicability, including whether the three phases remain similarly related across different HAII contexts.

Finally, although the sample was diverse, it was U.S.-based, and the studies relied on simulated rather than real-time interactions. The cultural, linguistic, and ecological generalizability of the findings therefore remains uncertain, and the English-only instruments require further validation across other linguistic and cultural contexts.

Supplementary material

Supplementary material is available at *Journal of Computer-Mediated Communication* online.

Data availability

The data underlying this article will be shared by the first author upon reasonable request.

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Conflicts of interest

None declared.

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